

D865

Secondary Physics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A physics teacher who writes a science blog sees a headline: “Study proves phone use causes poor grades.” The study surveyed 1,200 students and found that those who used phones more had lower grades. The blog has 5,000 subscribers. How should the teacher explain this to a general audience?

- A. Repeat the headline, since 1,200 students is a large sample.
- B. Say the study is worthless because surveys never give useful data.
- C. Use technical terms like “confounding” without explaining them.
- D. Explain that the survey shows an association, not proof of cause, and name other factors that could explain it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In a history-of-science lesson, a teacher with a replica of early lab apparatus explains that most alpha particles fired at thin gold foil passed straight through, but a few bounced back at large angles. The replica sits in a glass case. What conclusion did this evidence support?

- A. Positive charge is spread evenly through the atom.
- B. Gold atoms are solid, tightly packed spheres.
- C. The atom is mostly empty space with a small, dense, positively charged nucleus.
- D. Electrons travel in fixed orbits at set energies.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A city of 150,000, known for its riverfront murals, plans a new bridge. It must carry four lanes of traffic and last 75 years, within a fixed budget and without harming a protected fish habitat below. Which statement reflects sound engineering design?

- A. Treat the budget and habitat as goals to maximize.
- B. Define the criteria and constraints first, compare several designs, and choose the best tradeoff.
- C. Pick the strongest possible design regardless of budget.
- D. The first design that works is the best one.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student with a strong interest in weightlifting climbs a school stairwell 6.0 m high in 8.0 s. The student's mass is 60 kg. The stairwell was painted over the summer. What average power does the student develop? ($g = 9.8 \text{ m/s}^2$)

- A. 28,224 W
- B. About 441 W
- C. 3,528 W
- D. 45 W

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A 1,200 W motor runs for 30 s. How much work does it do?

- A. 36,000 J
- B. 1,200 J
- C. 36,000 W
- D. 40 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A gas in a cylinder, in a lab stocked with new glassware, absorbs 500 J of heat and does 200 J of work pushing a piston outward. The cylinder is 25 cm tall. What is the change in the gas's internal energy?

- A. +500 J
- B. -300 J
- C. +300 J
- D. +700 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A gas is compressed: 400 J of work is done on it while it releases 150 J of heat. What is the change in its internal energy?

- A. +250 J
- B. +550 J
- C. -250 J
- D. 0 J, since the gas both gains and loses energy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A kitchen circuit protected by a 20 A breaker has a toaster drawing 10 A. The homeowner plugs a 9 A kettle into the same circuit, which is wired in parallel. The kitchen was remodeled in 2021. What happens to the total current, and why?

- A. The breaker always trips when a second appliance is added.
- B. It rises to 19 A, because adding a parallel branch lowers total resistance; each appliance still gets full voltage.
- C. It stays at 10 A, split between the two.
- D. It drops, because adding a load raises resistance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Two $6\ \Omega$ resistors are connected in parallel across a 12 V battery. What is the total current?

- A. 0.25 A
- B. 1 A
- C. 2 A
- D. 4 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

For an engineering challenge, a teacher gives teams an iron nail, a 1.5 V battery, and insulated wire from a hardware-store kit. One team's electromagnet picks up 5 paper clips. The kit's box is bright orange. Which change will most increase the number of clips lifted?

- A. Wrap more turns of wire around the nail.
- B. Use wire with thicker insulation.
- C. Replace the iron nail with a plastic rod.
- D. Nothing about the coil matters; only the battery does.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A planetarium educator, preparing a show with a new dome projector, explains that the Sun fuses hydrogen into helium. Four hydrogen nuclei have slightly more mass than the helium nucleus they form. The show lasts 35 minutes. Where does the Sun's energy come from?

- A. Mass is created in fusion, which absorbs energy.
- B. Splitting heavy nuclei (fission).
- C. Hydrogen burning chemically with oxygen.
- D. The missing mass is converted into energy, $E = mc^2$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

12. [TRANSFER — reversed stem / different variable]

In a nuclear reaction, 0.001 kg of mass is converted into energy. How much energy is released? ($c = 3 \times 10^8$ m/s)

- A. About 1.1×10^{-20} J
- B. 3×10^5 J
- C. 9×10^{13} J
- D. 9×10^{16} J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A physics teacher who writes a science blog sees a headline: “Study proves phone use causes poor grades.” The study surveyed 1,200 students and found that those who used phones more had lower grades. ~~The blog has 5,000 subscribers.~~ How should the teacher explain this to a general audience?

Target word: accurate communication **Decides:** A survey shows an association.

A	Wrong stem number	Sample size does not turn a correlation into cause.
B	Absolute language	Surveys are useful; they just cannot prove cause alone.
C	Adjacent category	Accurate but unclear; communication must fit the audience.
D	Correct	Survey data cannot isolate cause; honest communication states limits.

2. Answer C Concept 2

In a history-of-science lesson, a teacher ~~with a replica of early lab apparatus~~ explains that most alpha particles fired at thin gold foil passed straight through, but a few bounced back at large angles. ~~The replica sits in a glass case.~~ What conclusion did this evidence support?

Target word: conclusion from the gold-foil results **Decides:** Most pass; a few bounce back.

A	Confusion set	That is the plum-pudding model the experiment overturned.
B	Inverted operation	If so, most particles would have bounced, not passed through.
C	Correct	Rare sharp deflections require concentrated positive mass.
D	Adjacent category	That is the later Bohr model, based on spectra, not this experiment.

3. Answer B Concept 3

A city of 150,000, ~~known for its riverfront murals~~, plans a new bridge. It must carry four lanes of traffic and last 75 years, within a fixed budget and without harming a protected fish habitat below. Which statement reflects sound engineering design?

Target word: design approach **Decides:** Criteria and constraints set first; tradeoffs compared.

A	Confusion set	Those are constraints (limits), not criteria to maximize.
B	Correct	No design maximizes every criterion within every limit.
C	Dropped element	Drops the constraints.
D	Absolute language	Comparing options usually finds better tradeoffs.

4. Answer B Concept 4

A student ~~with a strong interest in weightlifting~~ climbs a school stairwell 6.0 m high in 8.0 s. The student’s mass is 60 kg. ~~The stairwell was painted over the summer.~~ What average power does the student develop? ($g = 9.8 \text{ m/s}^2$)

Target word: average power **Decides:** $P = mgh \div t$.

A	Inverted operation	Multiplies by time instead of dividing.
B	Correct	$60 \times 9.8 \times 6.0 = 3,528 \text{ J}$; $3,528 \div 8.0 = 441 \text{ W}$.
C	Confusion set	Reports the work in joules as power.
D	Dropped element	Drops g .

5. Answer A Concept 4 • transfer

A 1,200 W motor runs for 30 s. How much work does it do?

Target word: work from power and time **Decides:** $W = P \times t$.

A	Correct	$1,200 \times 30 = 36,000 \text{ J}$.
B	Dropped element	Drops the time.
C	Confusion set	Right number, wrong unit; work is in joules.
D	Inverted operation	Divides power by time.

6. Answer C Concept 5

A gas in a cylinder, ~~in a lab stocked with new glassware~~, absorbs 500 J of heat and does 200 J of work pushing a piston outward. ~~The cylinder is 25 cm tall.~~ What is the change in the gas's internal energy?

Target word: ΔU **Decides:** $\Delta U = Q - W$.

A	Dropped element	Drops the work term.
B	Confusion set	Swaps the sign convention.
C	Correct	$500 - 200 = +300 \text{ J}$.
D	Inverted operation	Adds the work the gas does instead of subtracting it.

7. Answer A Concept 5 • transfer

A gas is compressed: 400 J of work is done on it while it releases 150 J of heat. What is the change in its internal energy?

Target word: ΔU with work done on the gas **Decides:** $Q = -150 \text{ J}$; $W(\text{by gas}) = -400 \text{ J}$.

A	Correct	$\Delta U = -150 - (-400) = +250 \text{ J}$.
B	Confusion set	Treats both as energy added.
C	Inverted operation	Flips the sign: work done on the gas adds energy.
D	Absolute language	Gains and losses are unequal here.

8. Answer B Concept 6

A kitchen circuit protected by a 20 A breaker has a toaster drawing 10 A. The homeowner plugs a 9 A kettle into the same circuit, which is wired in parallel. ~~The kitchen was remodeled in 2021.~~ What happens to the total current, and why?

Target word: effect of adding a parallel load **Decides:** Each branch gets full voltage.

A	Absolute language	19 A is under the 20 A rating.
B	Correct	Branch currents add at the panel.
C	Confusion set	Series thinking; parallel branches each draw their own current.
D	Inverted operation	Parallel branches lower total resistance.

9. Answer D Concept 6 · transfer

Two $6\ \Omega$ resistors are connected in parallel across a 12 V battery. What is the total current?

Target word: total current in parallel **Decides:** $R_{eq} = 3\ \Omega$.

A	Inverted operation	Divides R by V.
B	Confusion set	Adds the resistors as if in series ($12\ \Omega$).
C	Dropped element	Counts only one branch.
D	Correct	$12 \div 3 = 4\ \text{A}$ (2 A per branch).

10. Answer A Concept 7

For an engineering challenge, a teacher gives teams an iron nail, a 1.5 V battery, and insulated wire ~~from a hardware store kit~~. One team's electromagnet picks up 5 paper clips. ~~The kit's box is bright orange.~~ Which change will most increase the number of clips lifted?

Target word: how to strengthen the electromagnet **Decides:** Turns, current, core.

A	Correct	More turns at the same current make a stronger field.
B	Adjacent category	Insulation prevents shorts but does not strengthen the field.
C	Inverted operation	Iron concentrates the field; plastic weakens it.
D	Absolute language	Turns and core matter as much as current.

11. Answer D Concept 8

A planetarium educator, preparing a show ~~with a new dome projector~~, explains that the Sun fuses hydrogen into helium. Four hydrogen nuclei have slightly more mass than the helium nucleus they form. ~~The show lasts 35 minutes.~~ Where does the Sun's energy come from?

Target word: source of fusion energy **Decides:** Mass is lost in the reaction.

A	Inverted operation	Mass is lost, and energy is released.
B	Adjacent category	The Sun fuses light nuclei; it does not split heavy ones.
C	Confusion set	Chemical burning could not power the Sun for billions of years.
D	Correct	A tiny mass loss times c^2 is a huge energy.

12. Answer C Concept 8 · transfer

In a nuclear reaction, 0.001 kg of mass is converted into energy. How much energy is released? ($c = 3 \times 10^8\ \text{m/s}$)

Target word: energy from mass **Decides:** $E = mc^2$.

A	Inverted operation	Divides by c^2 instead of multiplying.
B	Dropped element	Drops the square on c.
C	Correct	$0.001 \times (3 \times 10^8)^2 = 0.001 \times 9 \times 10^{16} = 9 \times 10^{13}\ \text{J}$.
D	Wrong stem number	Uses 1 kg instead of 0.001 kg.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Communicating science accurately	Good science communication fits the audience, states what the evidence does and does not show, and avoids turning an association into proof of cause.	Repeating a “proves” headline or hiding behind jargon.	A media claim built on survey data.
2	Evidence changes models (gold-foil experiment)	Most alpha particles passed through gold foil but a few bounced back, showing the atom is mostly empty space with a small, dense, positive nucleus.	Crediting the result to the plum-pudding or orbit model.	Most particles pass, a few deflect sharply.
3	Engineering criteria, constraints, and tradeoffs	Engineers define criteria (what success means) and constraints (limits) first, compare several designs, and choose the best tradeoff; no design maximizes everything.	Maximizing one feature while ignoring limits.	A design problem listing goals and limits.
4	Work and power	Work = force $\times$ distance (lifting: mgh); power = work $\div$ time.	Reporting work as power or forgetting time.	A mass, a height, and a time.
5	First law of thermodynamics	$\Delta U = Q - W$, where Q is heat added to the system and W is work done by the system.	Adding the work the gas does, or flipping signs.	Heat and work given for a gas.
6	Parallel circuits	Each parallel branch gets the full voltage; adding a branch lowers total resistance and raises total current.	Thinking adding a branch raises resistance or splits a fixed current.	Appliances on one household circuit.
7	Electromagnets	A current makes a magnetic field; an electromagnet grows stronger with more turns, more current, and an iron core.	Thinking only the battery sets strength, or that any core works.	A coil-and-nail build with options to change.
8	Mass-energy in fusion	In fusion, the products have slightly less mass than the reactants; the missing mass is released as energy, $E = mc^2$.	Thinking the Sun burns chemically or runs on fission.	Reactant and product masses that differ slightly.

Course scope (public description): MAT Secondary Physics Curriculum This course offers a comprehensive review for high school physics teachers preparing for the Praxis exam. Explore major scientific discoveries, key engineering innovations, and their societal impacts. Understand ethical issues in scientific research and the principles of effective science communication. Delve into different forms of energy, laws of thermodynamics, and the role of energy and matter in the universe. Study Coulomb's Law, electric circuits, and electromagnetism. Learn how to integrate wave properties, sound waves, and the electromagnetic spectrum into your teaching practice. As this course is focused on Praxis prep, content is conceptual and algebra-based.